

PROBLEM SET 10

Show **ALL WORK** to get full/partial credit. Begin each problem on a new page, and clearly label each part of the problem. (**Max Score:** 20)

1) **(10 pts)** Note that the angular momentum seems to generate a repulsive potential term in Eq. (12.3.13).

(a) **(5 pts)** Identify this repulsive term $V_{\text{rep}}(\rho)$

(b) **(5 pts)** Calculate the gradient of this repulsive term (identified as the centrifugal force)

2) **(10 pts)** Consider a particle of mass μ constrained to move on a circle of radius a .

(a) **(5 pts)** Show that $H = \frac{L_z^2}{2\mu a^2}$

Hint: note that the particle only has kinetic energy, hence you can start with $H = P^2/2\mu$, where $P \rightarrow$ momentum operator in polar coordinates.

(b) **(5 pts)** Solve the eigenvalue problem of H to show that $E_m = \frac{\hbar^2 m^2}{2\mu a^2}$ and interpret the degeneracy (what is the degeneracy value?)